DEBUGGING REPORT

MIVA Level 300 Result Calculator System

1. INTRODUCTION

This report documents the debugging process carried out on a Python program designed to calculate the average score and grade of a **MIVA Level 300 student**.

The program initially contained multiple errors, including **syntax errors, runtime errors, and logical issues**. These errors were identified and resolved using debugging tools such as breakpoints, step execution, and variable inspection within the Visual Studio Code IDE.

2. PROGRAM OVERVIEW

The program performs the following functions:

- Accepts student scores
- Calculates the average score
- Assigns a grade
- Displays pass/fail status

However, the first version of the program contained intentional bugs for debugging practice.

3. BUGS ENCOUNTERED, IDENTIFICATION, AND FIXES

BUG 1: Syntax Error

Location:

```
if average > 50
```

Problem:

- Missing colon `:` at the end of the `if` statement
- Prevents the program from running

How It Was Identified:

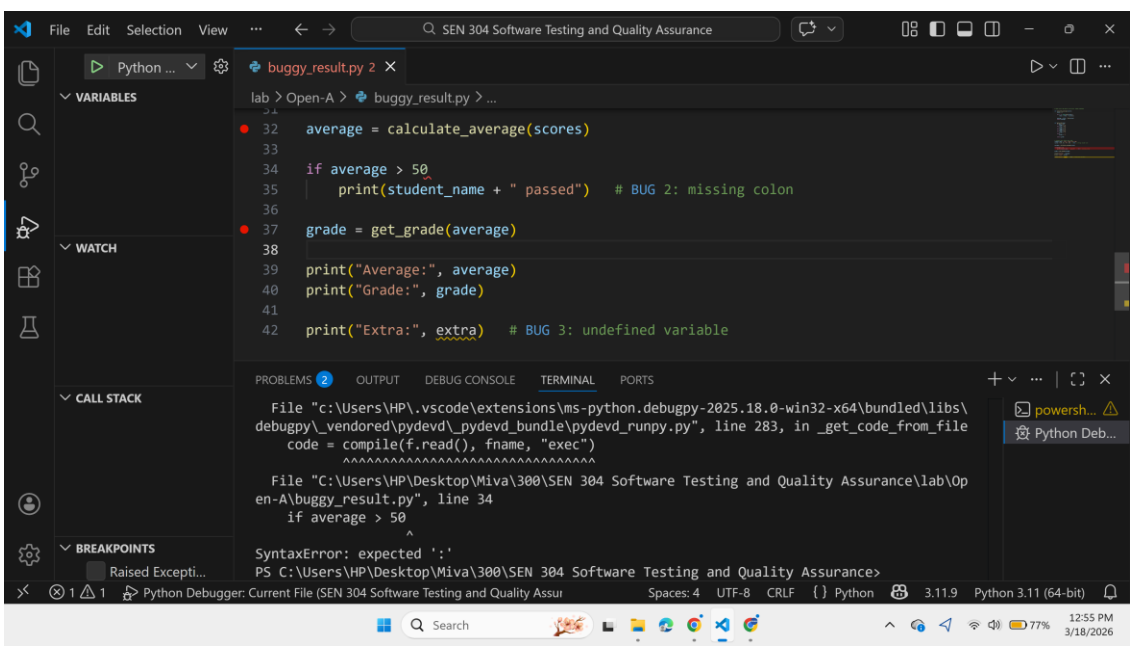
- When the program was executed, Python raised the error:

```
SyntaxError: invalid syntax
```

Fix:

```
if average > 50:
```

Screenshot:



BUG 2: Runtime Error (Type Error)

Location:

```
scores = [70, 65, "80", 90]
```

Problem:

- A string value `"80"` was included in a list of integers
- Caused failure during addition inside the loop

How It Was Identified:

- Breakpoint was set at:

```
average = calculate_average(scores)
```

- Used **Step Into (F11)** to enter the function
- Stepped through the loop using **Step Over (F10)**
- Error occurred at:

```
total = total + scores[i]
```

- Error message observed:

```
TypeError: unsupported operand type(s)
```

- Using **Watch Expressions**, it was observed that:

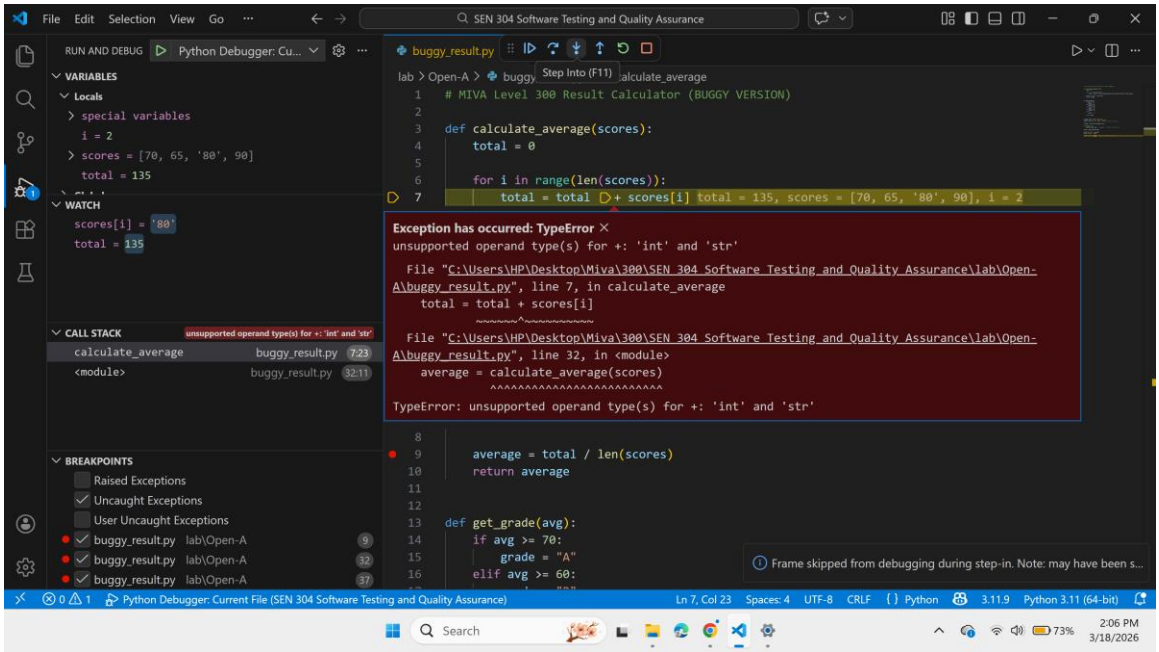
```
scores[i] = "80"
```

Fix:

Converted each score to a float:

```
total = total + float(scores[i])
```

Screenshot:



BUG 3: Undefined Variable (NameError)

Location:

```
print("Extra:", extra)
```

Problem:

- Variable `extra` was not defined anywhere in the program

How It Was Identified:

- When the program was executed after fixing previous bugs, it raised:

```
NameError: name 'extra' is not defined
```

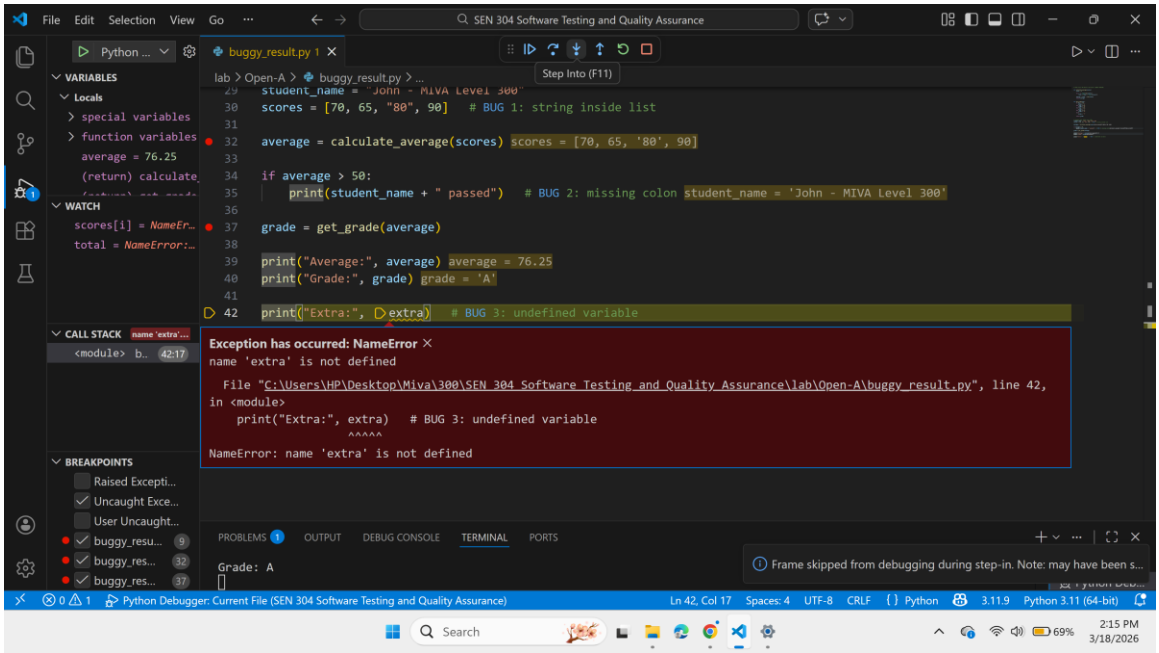
- The error location was clearly highlighted in the IDE

Fix:

Removed the unnecessary line:

```
# print("Extra:", extra)
```

Screenshot:



BUG 4: Logical Issue (Lack of Input Validation)

Problem:

- The program assumed all inputs were valid numeric values
- This could lead to runtime errors when invalid data is provided

How It Was Identified:

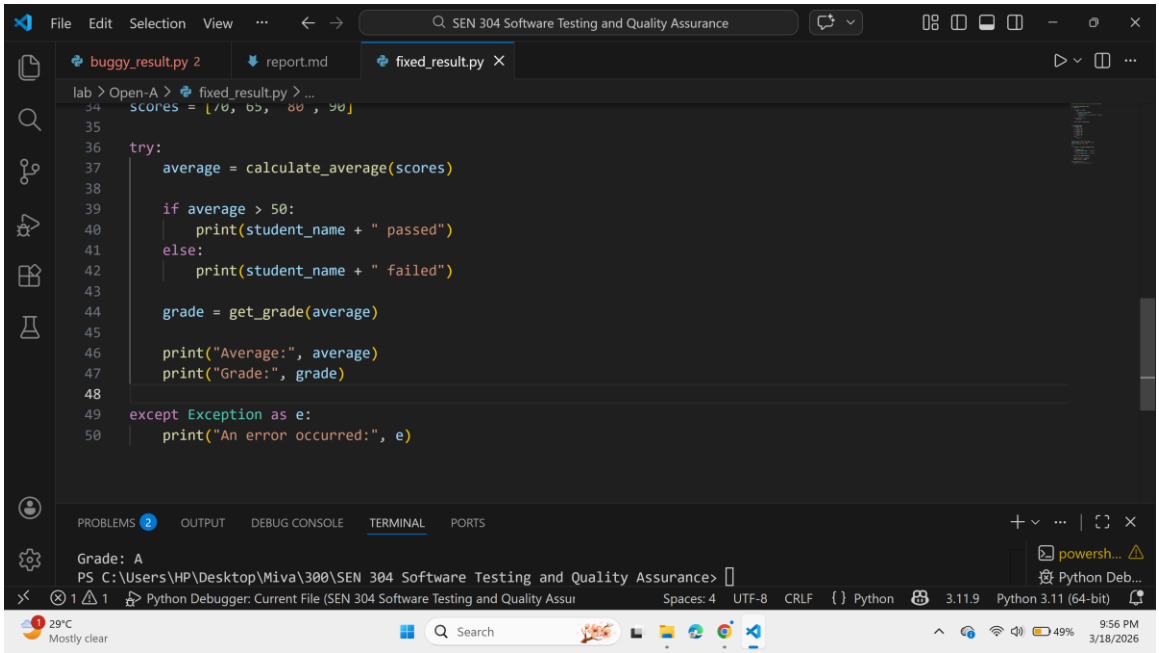
- During debugging, invalid data caused the program to crash
- Highlighted the need for safer data handling

Fix:

Improved code by converting values safely and ensuring proper calculations:

```
total = total + float(scores[i])
```

Screenshot:



4. DEBUGGING TOOLS USED

The following debugging tools were used in the IDE:

Breakpoints

- Set at key lines to pause execution

Step Execution

- **F11 (Step Into)**: Entered functions
- **F10 (Step Over)**: Executed code line by line

Watch Expressions

- Monitored variables such as:
  - `scores[i]`
  - `total`

Variable Inspection

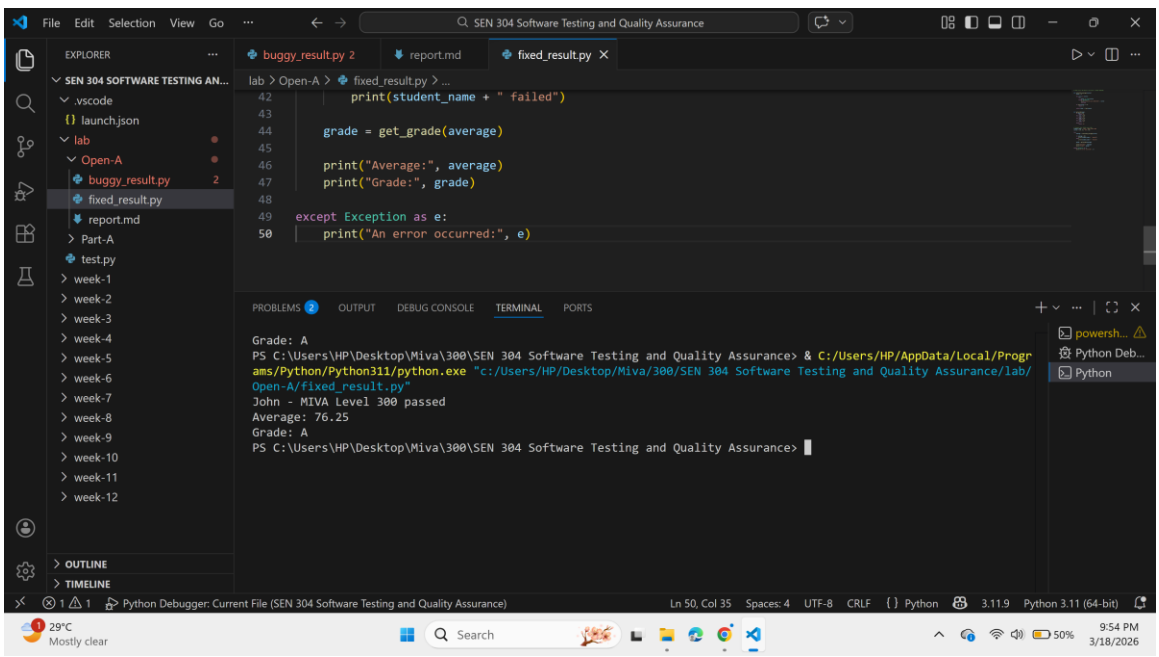
- Observed real-time values of variables during execution

5. FINAL PROGRAM OUTPUT

After fixing all errors, the program executed successfully and produced:

```
John - MIVA Level 300 passed
Average: 76.25
Grade: A
```

Screenshot:



6. CONCLUSION

The debugging process successfully identified and resolved all errors in the program.

Through the use of debugging tools in Vs code:

- Errors were traced systematically
- Program flow was clearly understood
- Code reliability was improved

This exercise demonstrates the importance of debugging in software development and highlights how tools like breakpoints and variable inspection can be effectively used to resolve issues.